

MODELLING AND DEMONSTRATION OF FERRANTI EFFECT

Assignment 4

220 kV, 50 Hz, 200 km Transmission System

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Objective

To model a 220 kV, 50 Hz, 200 km three-phase transmission line in MATLAB/Simulink, demonstrate the Ferranti effect under no-load/light-load conditions, and demonstrate a suitable mitigation method using a shunt reactor.

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1 Introduction

The Ferranti effect is the phenomenon in which the receiving-end voltage of a long transmission line becomes greater than the sending-end voltage when the line operates under no-load or light-load conditions.

$$V_R > V_S$$

The major cause is the shunt capacitance of the line. Even when the receiving end is open, the line capacitance draws a charging current. The interaction of this capacitive current with the series inductance produces a rise in the receiving-end voltage.

2 System Specifications

Table 1: Transmission line specifications

Parameter	Specified Value
Line-to-line voltage	220 kV RMS
Frequency	50 Hz
Line length	200 km
Series resistance	0.05 Ω /km/phase
Series inductance	1 mH/km/phase
Shunt capacitance	0.01 μ F/km/phase

3 Theoretical Analysis

For a 200 km line,

$$R = 0.05(200) = 10 \, \Omega, \quad L = 1 \times 10^{-3}(200) = 0.2 \, \text{H},$$

$$C = 0.01 \times 10^{-6}(200) = 2 \, \mu\text{F},$$

and

$$\omega = 2\pi(50) = 314.159 \, \text{rad/s}.$$

Therefore,

$$Z = R + j\omega L = 10 + j62.8319 \, \Omega,$$

$$Y = j\omega C = j0.0006283185 \, \text{S}.$$

For the nominal- π model,

$$A = D = 1 + \frac{ZY}{2}, \quad B = Z \left(1 + \frac{ZY}{6} \right), \quad C = Y \left(1 + \frac{ZY}{6} \right).$$

For no-load operation,

$$I_R = 0, \quad V_R = \frac{V_S}{A}.$$

For the specified system,

$$\boxed{V_R \approx 224.43 \, \text{kV}}.$$

The corresponding voltage rise is

$$\frac{224.43 - 220}{220} \times 100 = \boxed{2.0131\%}.$$

4 Mitigation of Ferranti Effect

A suitable method for reducing Ferranti-effect overvoltage is to connect a shunt reactor at the receiving end.

The line capacitance supplies capacitive reactive power, while the shunt reactor absorbs inductive reactive power.

$$Q_C + Q_L \approx 0$$

when the reactor is suitably tuned.

For the analytical benchmark,

$$\boxed{Q_{\text{reactor}} \approx 15.35 \, \text{MVar}}.$$

This restores the no-load receiving-end voltage to approximately

220 kV

.

5 Procedure

1. Set the three-phase source to 220 kV line-to-line RMS and 50 Hz.
2. Set the transmission-line length to 200 km.
3. Enter the specified resistance, inductance and capacitance values.
4. Leave the receiving end open for the no-load test.
5. Measure the receiving-end line-to-line voltage using the Line Voltage Sensor.
6. Convert the physical signal using the PS-Simulink Converter.
7. Calculate RMS using the RMS Measurement block.
8. Connect the RMS output to the Display.
9. Connect a branch to the Scope to observe the three receiving-end waveforms.
10. Run the simulation for 0.2 s.
11. Compare the receiving-end voltage with the 220 kV sending-end voltage.
12. Repeat for the light-load condition.
13. Add a receiving-end shunt reactor and verify the reduction of overvoltage.

6 MATLAB/Simulink Model

The Simulink model consists of a three-phase voltage source, Transmission Line (Three-Phase), Line Voltage Sensor (Three-Phase), PS-Simulink Converter, RMS Measurement, Display, Scope, Solver Configuration and Electrical Reference.

The receiving end is initially kept open for the no-load condition.

The measurement chain is

Line Voltage Sensor → PS-Simulink Converter → RMS Measurement → Display

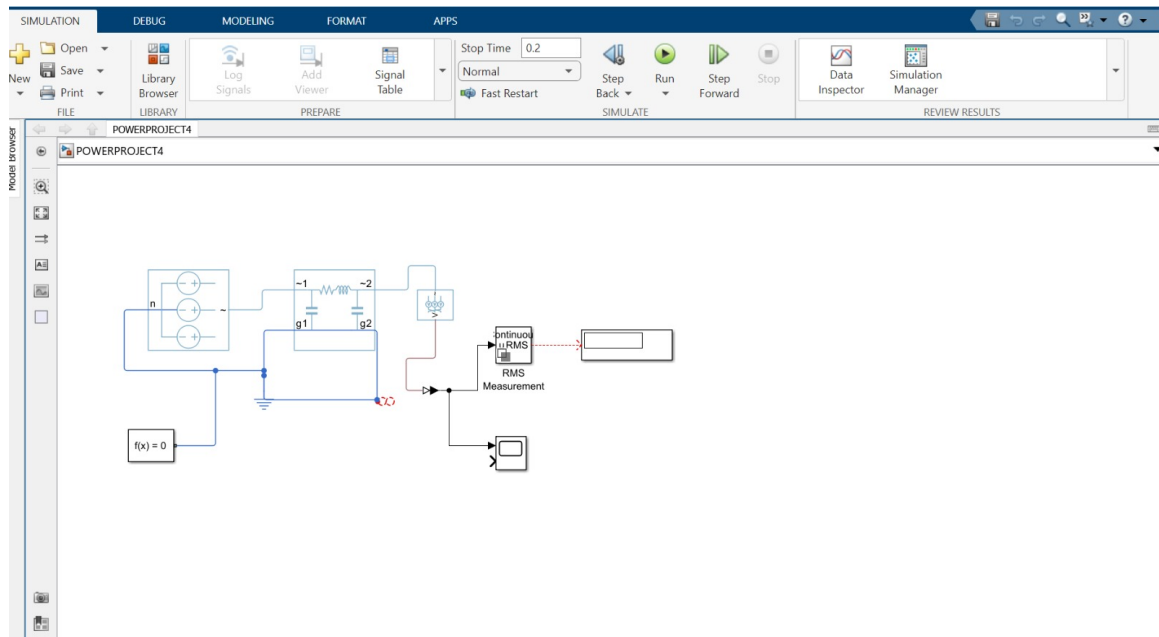


Figure 1: MATLAB/Simulink block diagram of the transmission system.

6.1 Simulation Settings

- Sending-end voltage = 220 kV line-to-line RMS
- Frequency = 50 Hz
- Line length = 200 km
- Resistance = $0.05 \Omega/\text{km}$
- Inductance = $1 \text{ mH}/\text{km}$
- Simulation stop time = 0.2 s

6.2 Receiving-End Waveforms

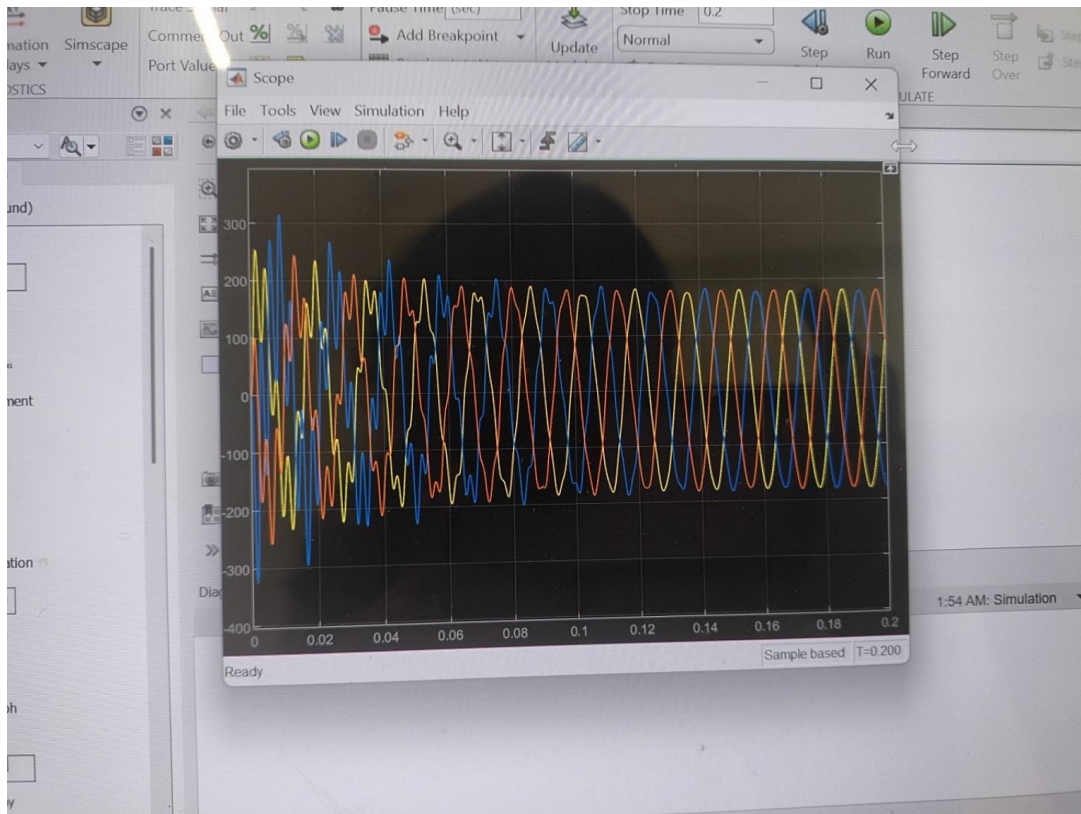


Figure 2: Receiving-end three-phase line-to-line voltage waveforms observed using the Scope.

The waveforms show three balanced sinusoidal receiving-end voltages after the initial startup transient.

7 MATLAB Code

The following MATLAB program performs the analytical nominal- π benchmark, light-load calculation, receiving-end shunt-reactor tuning, result generation, and voltage-versus-load plots.

```

1  %% Assignment 4 - Ferranti Effect
2  % 220 kV, 50 Hz, 200 km transmission line
3  % Analytical nominal- $\pi$  benchmark for Simulink verification
4
5  clear;
6  clc;
7  close all;
8
9  fprintf('=====\n');
10 fprintf(' Assignment 4: Ferranti Effect - 220 kV, 50 Hz, 200 km\n');
11 fprintf('=====\n\n');
```

```

12
13 %% INPUT DATA
14
15 VLL_kV = input('Line-line voltage in kV : ');
16 if isempty(VLL_kV)
17     VLL_kV = 220;
18 end
19
20 f = input('Frequency in Hz : ');
21 if isempty(f)
22     f = 50;
23 end
24
25 len_km = input('Line length in km : ');
26 if isempty(len_km)
27     len_km = 200;
28 end
29
30 R_km = input('Series resistance in ohm/km/phase : ');
31 if isempty(R_km)
32     R_km = 0.05;
33 end
34
35 L_km_mH = input('Series inductance in mH/km/phase : ');
36 if isempty(L_km_mH)
37     L_km_mH = 1;
38 end
39
40 C_km_uF = input('Shunt capacitance in uF/km/phase : ');
41 if isempty(C_km_uF)
42     C_km_uF = 0.01;
43 end
44
45 P_light_MW = input('Light-load active power in MW : ');
46 if isempty(P_light_MW)
47     P_light_MW = 50;
48 end
49
50 pf_light = input('Light-load lagging power factor : ');
51 if isempty(pf_light)
52     pf_light = 0.95;
53 end
54
55 %% LINE PARAMETERS
56
57 VLL = VLL_kV * 1e3;
58 w = 2 * pi * f;
59
60 R = R_km * len_km;
61 L = (L_km_mH * 1e-3) * len_km;
62 C = (C_km_uF * 1e-6) * len_km;
63
64 Vphs = VLL / sqrt(3);
65 Vs = Vphs;
66
67 %% NOMINAL-PI ABCD PARAMETERS
68
69 Z = R + 1j * w * L;
70 Y = 1j * w * C;

```

```

71
72 A    = 1 + Z * Y / 2;
73 B    = Z * (1 + Z * Y / 6);
74 C = Y * (1 + Z * Y / 6);
75 D    = A;
76
77 %% NO-LOAD CONDITION
78
79 Vr0 = Vs / A;
80
81 Ir0 = CabC * Vr0;
82 Is0 = A * Ir0 + CabC * Vr0;
83 S0  = 3 * Vs * conj(Is0);
84
85 Vrecv0_kV = abs(Vr0) * sqrt(3) / 1e3;
86
87 Ferranti_percent = ...
88     (Vrecv0_kV - VLL_kV) / VLL_kV * 100;
89
90 %% LIGHT-LOAD CONDITION
91
92 P = P_light_MW * 1e6;
93 Q = P * tan(acos(pf_light));
94
95 Vr = Vphs;
96
97 for k = 1:1000
98
99     Ir = conj((P + 1j * Q) / (3 * Vr));
100    Vr_new = (Vs - B * Ir) / A;
101
102    if abs(Vr_new - Vr) < 1e-8
103        Vr = Vr_new;
104        break;
105    end
106
107    Vr = Vr_new;
108
109 end
110
111 Ir_light = conj((P + 1j * Q) / (3 * Vr));
112
113 Is_light = A * Ir_light + CabC * Vr;
114 SL        = 3 * Vs * conj(Is_light);
115
116 VrecvL_kV = abs(Vr) * sqrt(3) / 1e3;
117
118 Light_deviation_percent = ...
119     (VrecvL_kV - VLL_kV) / VLL_kV * 100;
120
121 %% SHUNT-REACTOR MITIGATION
122
123 % Tune the receiving-end reactor so that the no-load
124 % receiving-end voltage is restored to nominal.
125
126 target = VLL;
127
128 fun = @(Br) abs( ...
129     Vs / (A + B * (-1j * Br)) ...

```

```

130     ) * sqrt(3) - target;
131
132 Br = fzero(fun, [0 0.005]);
133
134 Yr = -1j * Br;
135 VrR = Vs / (A + B * Yr);
136
137 IrR = Yr * VrR;
138 IsR = A * IrR + Cabc * VrR;
139 SR = 3 * Vs * conj(IsR);
140
141 Qreactor_Mvar = ...
142     3 * (VLL / sqrt(3))^2 * Br / 1e6;
143
144 Lreactor_H = 1 / (w * Br);
145
146 VrecvR_kV = abs(VrR) * sqrt(3) / 1e3;
147
148 Reactor_deviation_percent = ...
149     (VrecvR_kV - VLL_kV) / VLL_kV * 100;
150
151 %% DISPLAY LINE PARAMETERS
152
153 fprintf('\n--- LINE PARAMETERS ---\n');
154
155 fprintf('R = %.4f ohm/phase\n', R);
156 fprintf('L = %.4f H/phase\n', L);
157 fprintf('C = %.6e F/phase\n', C);
158 fprintf('Z = %.4f + j%.4f ohm/phase\n', real(Z), imag(Z));
159 fprintf('Y = j%.8f S/phase\n', imag(Y));
160
161 %% DISPLAY NO-LOAD RESULTS
162
163 fprintf('\n--- NO LOAD ---\n');
164
165 fprintf('Receiving V = %.3f kV L-L\n', Vrecv0_kV);
166 fprintf('Ferranti voltage rise = %.3f %%\n', Ferranti_percent);
167 fprintf('Receiving current = %.3f A\n', abs(Ir0));
168
169 %% DISPLAY LIGHT-LOAD RESULTS
170
171 fprintf('\n--- LIGHT LOAD ---\n');
172
173 fprintf('P = %.2f MW, pf = %.3f lagging\n', ...
174     P_light_MW, pf_light);
175
176 fprintf('Qload = %.3f MVar\n', Q / 1e6);
177 fprintf('Receiving V = %.3f kV L-L\n', VrecvL_kV);
178 fprintf('Deviation = %.3f %%\n', Light_deviation_percent);
179
180 %% DISPLAY MITIGATION RESULTS
181
182 fprintf('\n--- MITIGATION ---\n');
183
184 fprintf('Reactor rating = %.3f MVar at 220 kV\n', Qreactor_Mvar);
185 fprintf('Equivalent reactor L = %.4f H/phase\n', Lreactor_H);
186 fprintf('Compensated receiving V = %.3f kV L-L\n', VrecvR_kV);
187
188 %% RESULTS TABLE

```

```

189
190 Case = {
191     'No load'
192     'Light load'
193     'No load + reactor'
194 };
195
196 Vrecv_kV = [
197     Vrecv0_kV
198     VrecvL_kV
199     VrecvR_kV
200 ];
201
202 Deviation_pct = [
203     Ferranti_percent
204     Light_deviation_percent
205     Reactor_deviation_percent
206 ];
207
208 disp(table(Case, Vrecv_kV, Deviation_pct));
209
210 %% RECEIVING VOLTAGE VS LOAD
211
212 Pscan_MW = 0:5:100;
213 Vscan_kV = zeros(size(Pscan_MW));
214
215 for n = 1:numel(Pscan_MW)
216
217     if Pscan_MW(n) == 0
218
219         Vscan_kV(n) = Vrecv0_kV;
220
221     else
222
223         Pn = Pscan_MW(n) * 1e6;
224         Qn = Pn * tan(acos(pf_light));
225         Vrn = Vphs;
226
227         for k = 1:1000
228
229             In = conj((Pn + 1j * Qn) / (3 * Vrn));
230             Vnew = (Vs - B * In) / A;
231
232             if abs(Vnew - Vrn) < 1e-8
233                 Vrn = Vnew;
234                 break;
235             end
236
237             Vrn = Vnew;
238
239         end
240
241         Vscan_kV(n) = abs(Vrn) * sqrt(3) / 1e3;
242
243     end
244 end
245
246 %% PLOT 1: VOLTAGE VS LOAD
247

```

```
248 figure('Color', 'w');
249
250 plot(Pscan_MW, Vscan_kV, 'LineWidth', 1.8);
251 hold on;
252
253 yline(VLL_kV, '--', '220 kV');
254
255 grid on;
256
257 xlabel('Receiving-end active power (MW)');
258 ylabel('Receiving-end voltage (kV L-L)');
259
260 title('Ferranti Effect: Receiving-End Voltage vs Load');
261
262 legend( ...
263     'Calculated voltage', ...
264     'Nominal voltage', ...
265     'Location', 'best');
266
267 %% PLOT 2: SHUNT-REACTOR MITIGATION
268
269 figure('Color', 'w');
270
271 bar(categorical(Case), Vrecv_kV);
272
273 grid on;
274
275 ylabel('Receiving-end voltage (kV L-L)');
276
277 title('Ferranti Effect and Shunt-Reactor Mitigation');
```

Listing 1: MATLAB program for Ferranti-effect analysis

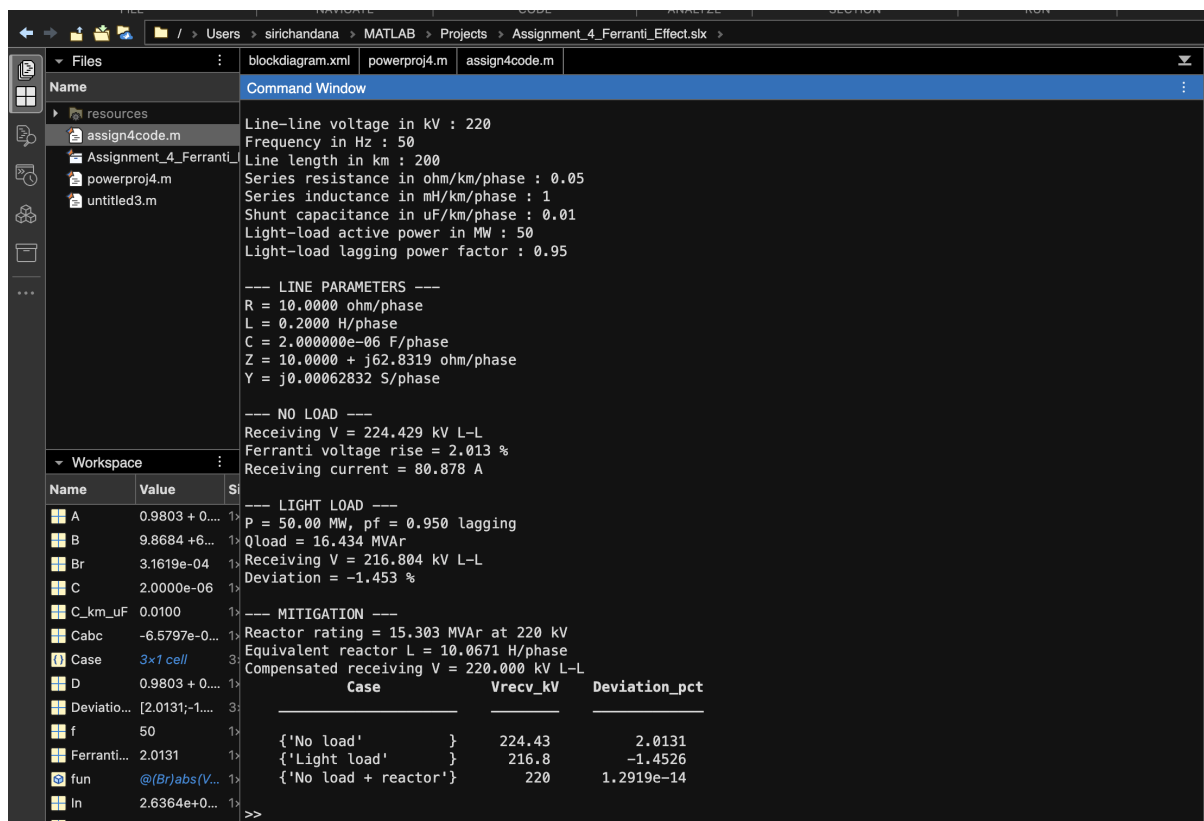


Figure 3: MATLAB Command Window output for the three operating cases.

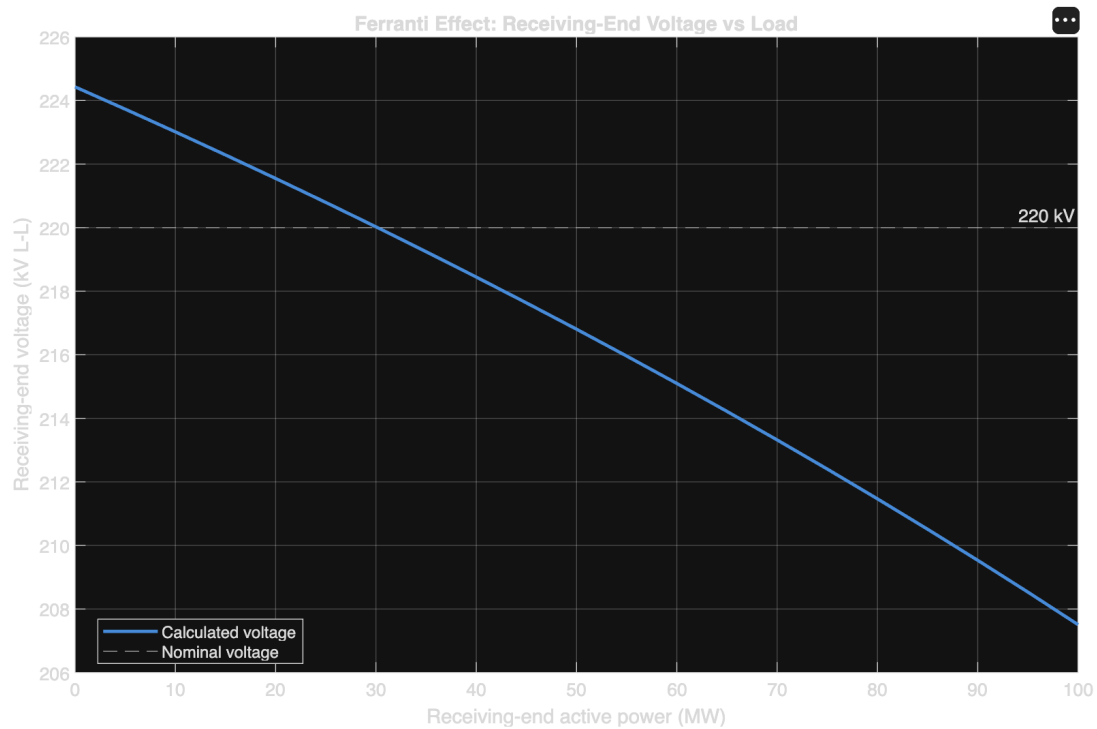


Figure 4: plot1

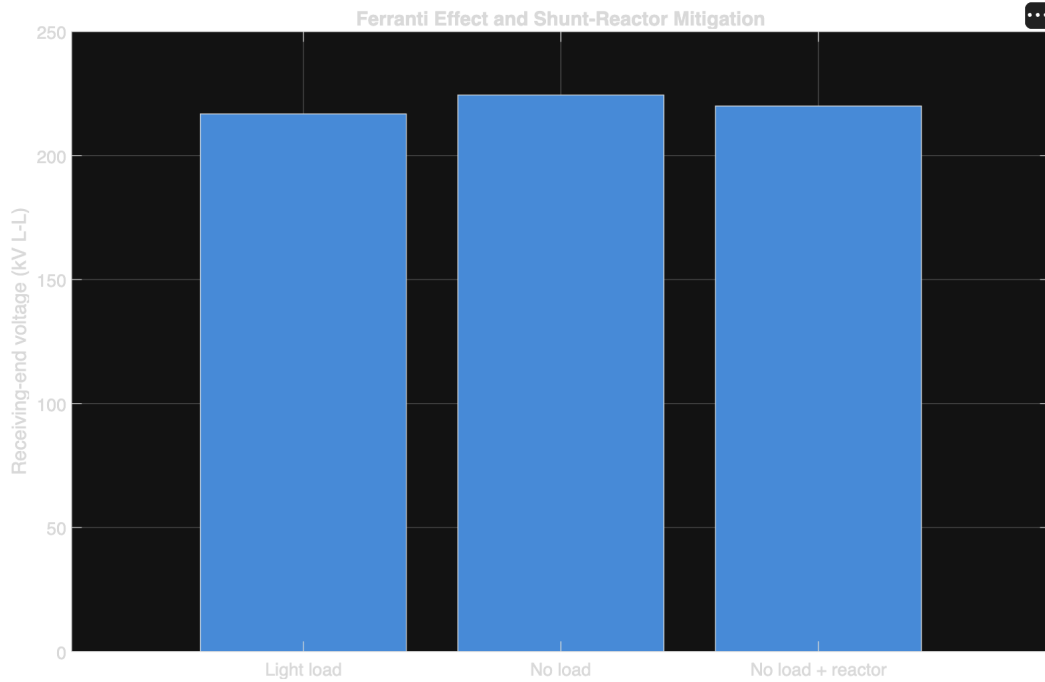


Figure 5: plot2

For the no-load case,

$$V_R = 224.43 \text{ kV}, \quad V_S = 220 \text{ kV},$$

so

$$V_R - V_S = 4.43 \text{ kV}.$$

The corresponding rise is

$$\frac{4.43}{220} \times 100 = 2.0131\%.$$

Thus,

$$V_R > V_S$$

which clearly demonstrates the Ferranti effect.

For the 50 MW, 0.95 lagging power-factor light-load case, the receiving-end voltage is 216.84 kV. The load current produces a voltage drop across the line impedance, reducing the no-load overvoltage.

With the tuned shunt reactor, the receiving-end voltage is restored to approximately 220 kV.

8 Conclusion

The 220 kV, 50 Hz, 200 km transmission line demonstrates the Ferranti effect under no-load conditions. The analytical receiving-end voltage is approximately

$$224.43 \text{ kV},$$

which is about

$$2.0131\%$$

above the 220 kV sending-end voltage.

The light-load case gives 216.84 kV, showing that loading reduces the no-load voltage rise. With a suitably sized receiving-end shunt reactor, the voltage is restored close to 220 kV. Therefore, shunt-reactor compensation is an effective method for controlling Ferranti-effect overvoltage in long transmission lines.

9 GITHUB

<https://github.com/K-SiriChandana/POWER-SYSTEMS-PROJECT-4>

Contains Matlab Simulink .slx file, Matlab Code .m file , ScreenShots of Result, Block Diagram, Report.